

Republic of Tunisia

Ministry of Higher Education and Scientific Research

National School of Engineers of Sousse

SEMESTER PROJECT REPORT ENTITLED:

BEM system — Group 4 :Communication & System Integration

From System Context to Simulink Implementation

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1. Introduction

1.1 Project Overview

This report documents the complete systems engineering lifecycle of Group 4 within the Battery Energy Management (BEM) System project at ENISO (École Nationale d'Ingénieurs de Sousse), carried out as part of the Automotive Software Engineering curriculum. The broader BEM project spans four groups — each responsible for a distinct functional domain — working collaboratively on a shared Model-Based Systems Engineering (MBSE) artifact representing a Tesla-like urban electric vehicle equipped with a 78.1 kWh, 96s46p lithium-ion battery pack.

Group 4 is responsible for the Communication and System Integration domain. This domain is the interface layer of the entire BEM architecture: it does not sense voltages, estimate states of charge, or physically actuate contactors. Instead, it ensures that every piece of critical battery data produced by Groups 1, 2, and 3 reaches the right external system — the Vehicle Control Unit (VCU), the charging infrastructure, the driver's dashboard, and the diagnostic tooling — in the right format, at the right time, with the right safety guarantees. Without this integration layer, the battery could be perfectly monitored internally while remaining completely invisible to the vehicle ecosystem surrounding it.

1.2 Role of Group 4 :

Group 4 is responsible for the Communication and System Integration domain. This domain acts as the nervous system of the BEM system - it does not directly measure cell voltages (Group 1), estimate state of charge (Group 2), or physically protect the battery (Group 3). Instead, it connects all internal BEM sub-functions to the external vehicle ecosystem and ensures that the right data reaches the right ECU at the right time in the right format. The main responsibilities of Group 4 are:

- CAN bus transmission and reception (500 kbps, CAN 2.0B)
- VCU interface management - SOC, voltage, current, temperature, State of Power to the Vehicle Control Unit
- Charging interface control - voltage and current negotiation with the EVSE over CAN
- UDS diagnostics and DTC logging per ISO 14229 and OBD-II
- BEM operating mode finite state machine - 6 modes with ASIL-C safety constraints

1.4 Target Vehicle and Battery Pack

The reference vehicle is a Tesla-like urban electric vehicle. The battery pack parameters that govern all Group 4 design decisions are summarized below.

Parameter	Value
Configuration	96 cells in series × 46 cells in parallel (4,416 cells total)
Nominal cell voltage	3.7 V
Nominal pack voltage	~355 V
Total capacity	~220 Ah
Total energy	~78.1 kWh
Cell chemistry	NMC (Nickel Manganese Cobalt)
Operating temperature	−40 °C to +85 °C
Safety standard	ISO 26262 ASIL-C

2. System Context and External Interfaces

2.1 BEM Position in the Vehicle Architecture

The vehicle employs a distributed electronic architecture in which all major ECUs communicate over a shared CAN bus network. The BEM occupies the central position in the energy management chain, acting as the authoritative source of battery state information for all other vehicle systems. The key external systems interfacing with the BEM are:

External System	Type	Interaction with BEM
Vehicle Control Unit (VCU)	Master ECU	Receives SOC, pack voltage, current, SoP; sends power requests
Electric Motor / Inverter	Powertrain	BEM limits discharge current to protect motor from overcurrent
Thermal Management System	Thermal ECU	BEM requests cooling or heating based on real-time cell temperature
Dashboard / Instrument Cluster	HMI	Receives SOC %, range estimate, warnings, and alert signals from BEM
Charging System (EVSE)	External Charger	BEM negotiates voltage and current limits for AC and DC charging
Diagnostic Tester (OBD-II)	Service Tool	BEM responds to UDS ISO 14229 requests and provides DTC data

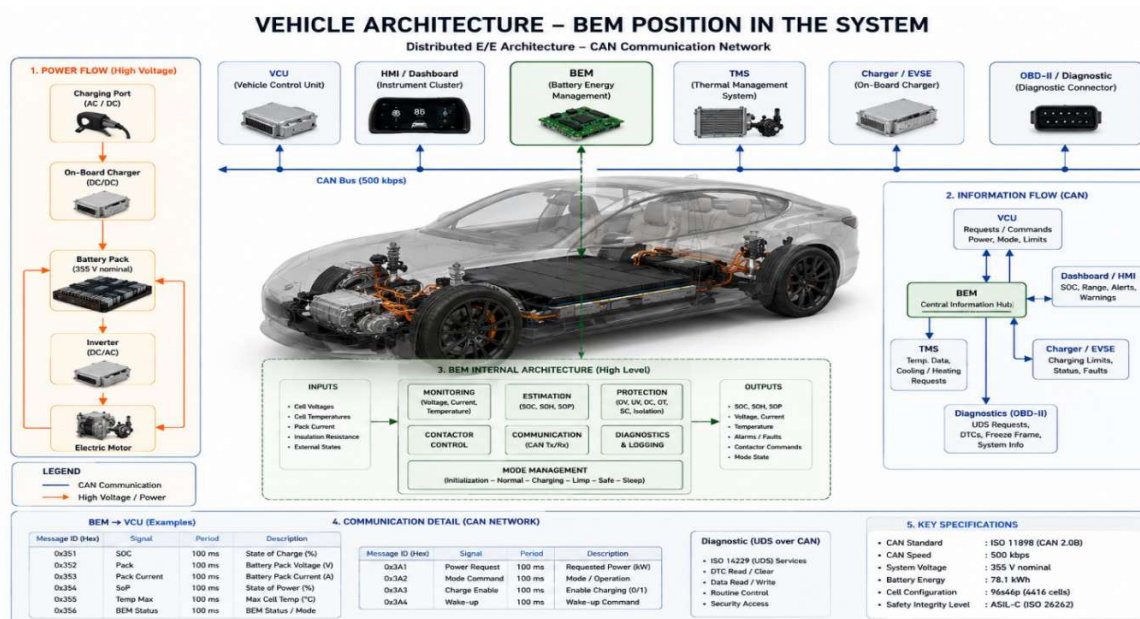
2.2 BEM Scope Boundaries

To avoid ambiguity in the multi-group project, explicit scope boundaries were established. The BEM is responsible for: cell voltage and temperature monitoring, state estimation (SOC/SOH/SoP), safety protection, contactor control, CAN communication, operating mode management, diagnostic logging, and charging management. The BEM is explicitly not responsible for: propulsion control algorithms, regenerative braking strategy, cabin climate comfort, or high-voltage cabling outside the battery enclosure.

2.3 High-Level BEM Functions

At the top level, the BEM performs seven distinct functions that establish the context for Group 4's integration responsibilities:

- Monitor — acquire cell voltage, current, and temperature data in real time (Group 1).
- Estimate — compute SOC, SOH, and State of Power (SoP) from measured data (Group 2).
- Protect — detect and react to fault conditions including overvoltage, thermal events, and short circuits (Group 3).
- Control Contactors — open and close main and pre-charge contactors under safe conditions (Group 3).
- Communicate — transmit BEM state over CAN to VCU, dashboard, charger, and diagnostics (Group 4).
- Diagnose — log DTCs, respond to UDS requests from service technicians (Group 4).
- Manage Operating Modes — coordinate Initialization, Normal, Charging, Limp, Safe State, and Sleep modes (Group 4).



3. Customer Needs and System Requirements

3.1 Customer Needs Analysis

The customer — the vehicle OEM — expressed operational needs at the vehicle level. Group 4 interpreted and translated the following customer needs into communication and integration requirements:

CN ID	Customer Need	Group 4 Interpretation
CN-013/014	Charging infrastructure compatibility	BEM must communicate charging parameters over CAN to EVSE (AC and DC)
CN-015/016	CAN communication at 500 kbps	BEM must transmit critical frames every 10 ms at 500 kbps, CAN 2.0B
CN-019	Accurate driver information	BEM must broadcast SOC, range, and warnings to dashboard via CAN
CN-009	React to dangerous situations	Critical fault reaction ≤ 100 ms; non-critical ≤ 500 ms
CN-024	Professional service support	BEM must support UDS ISO 14229 with $\geq 95\%$ DTC coverage
CN-027	Diagnostics for technicians	BEM must log DTCs with freeze frames accessible via OBD-II
CN-001/012	System availability $> 99.9\%$	BEM must wake from Sleep within 500 ms on CAN NM or KL15 signal

3.2 Group 4 Primary System Requirements

Nine system requirements were derived from the customer needs analysis and formalized in the Requirements Traceability Matrix (Proposed_RTM.xlsx). These requirements are owned by Group 4 and form the engineering contract for all subsequent design work:

Req ID	Title	Description	Category	Priority
BMS-SYS-019	DTC Logging	BMS shall log all faults with conditions and DTC code per OBD-II standard	Functional	High
BMS-SYS-020	Fault Reaction Time	Critical faults ≤ 100 ms; non-critical ≤ 500 ms	Non-Functional	Critical
BMS-SYS-021	CAN Communication	CAN 2.0B at 500 kbps; 10 ms cycle for critical messages	Functional	High

BMS-SYS-022	VCU Interface	BMS shall transmit SOC, voltage, current, temperature, SoP to VCU	Functional	High
BMS-SYS-023	Charging Interface	BMS shall control charging parameters (V, I) via CAN with EVSE	Functional	High
BMS-SYS-024	Diagnostic Interface	BMS shall support UDS protocol over CAN per ISO 14229	Functional	High
BMS-SYS-025	Operating Modes	BMS shall implement: Init, Normal, Charging, Limp, Safe, Sleep	Functional	High
BMS-SYS-026	Wake-up Management	BMS shall wake from Sleep within 500 ms on CAN NM or KL15	Non-Functional	Medium
BMS-SYS-027	Data Logging	BMS shall log SOC, V, I, T at 1 Hz to flash memory	Functional	Medium

3.3 Shared Non-Functional Requirements

Six non-functional requirements apply across all groups and constrain the entire BEM system design:

Req ID	Title	Constraint
BMS-SYS-028	Operating Temperature	−40 °C to +85 °C junction temperature
BMS-SYS-029	Operating Voltage	200 V to 800 V (±10% transients)
BMS-SYS-030	EMC Compliance	ISO 11452 (immunity) + ISO 7637 (conducted transients)
BMS-SYS-031	Safety Integrity	ISO 26262 ASIL-C for all safety-relevant functions
BMS-SYS-032	Availability	≥ 99.9% availability; MTTF ≥ 100,000 hours
BMS-SYS-033	Response Time Budget	Computation cycle ≤ 10 ms for all algorithms

4. Functional Architecture — SADT/IDEF0 Decomposition

4.1 Methodology

SADT (Structured Analysis and Design Technique), implemented through the IDEF0 notation, is a systems engineering methodology for modeling the functional architecture of complex systems. Each function is represented as a box with four types of interface arrows following the ICOM convention: Inputs (left — data consumed by the function), Outputs (right — data produced), Controls (top — standards, regulations, and constraints governing execution), and Mechanisms (bottom — resources that perform the function). Functions are decomposed hierarchically from a top-level A0 context diagram through successive levels of detail.

4.2 A0 — Top-Level Context Diagram

At the A0 level, the entire Group 4 scope is represented as a single function: Manage BEM Communication and System Integration. The full ICOM for this top-level context is as follows:

ICOM	Content
INPUT	SOC / SOH / SoP — state estimates from Group 2 (BMS_EstimationModule)
INPUT	Cell V / I / T_max / T_min — measurement data from Group 1 (BMS_MonitoringModule)
INPUT	Fault Events + Severity — fault classification from Group 3 (BMS_SafetyModule)
INPUT	Wake CAN Frame / KL15 — external wake trigger from CAN NM or ignition
OUTPUT	CAN Frames to VCU — SOC, V, I, T, SoP at 10 ms cycle
OUTPUT	Charging Parameters to EVSE — voltage target, current limit, charge enable
OUTPUT	UDS Response / DTC to Tester — ISO 14229 responses, DTC + freeze frames
OUTPUT	Mode Status / Warnings to HMI — active mode, alerts to dashboard
CONTROL	ISO 26262 ASIL-C — safety constraints on fault reaction timing
CONTROL	ISO 11898 / ISO 14229 — CAN standard and UDS protocol
CONTROL	OEM DBC Message Schedule — message IDs, periods, signal encoding
MECHANISM	SPC5748G MCU — dual-CAN microcontroller hardware
MECHANISM	AUTOSAR COM/PDUR/CanIf — software communication stack

MECHANISM	Dcm + Dem (UDS/DTC) — AUTOSAR diagnostic service modules
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4.3 A1 — CAN Bus Transmission and Reception

This sub-function handles all CAN bus TX/RX at 500 kbps, including periodic and event-driven frame scheduling, bus-off recovery, and raw frame parsing from VCU and EVSE. It directly satisfies BMS-SYS-021 and BMS-SYS-033.

ICOM	Content
INPUTS	BMS internal data (SOC, V, I, T, SoP) from G1/G2/G3 · Incoming CAN frames from VCU (CAN ID 0x200) · Incoming CAN frames from EVSE
OUTPUTS	Periodic CAN frames (BMS_Status 10 ms, BMS_Energy 100 ms) · Parsed VCU power request to A2 and A5 · Parsed EVSE capability to A3
CONTROLS	ISO 11898-1 · OEM DBC message cycle table · ASIL-C SW timing budget (≤ 10 ms, BMS-SYS-033)
MECHANISMS	SPC5748G dual CAN controller · AUTOSAR CanIf + PDUR + COM · Hardware TX/RX FIFO

4.4 A2 — VCU Interface Management

This sub-function transmits the complete BEM state to the VCU and receives power arbitration requests. All signals are encoded per the OEM DBC with AUTOSAR E2E CRC protection. Satisfies BMS-SYS-022.

ICOM	Content
INPUTS	SOC [%], V_pack [V], I_pack [A], SoP_charge [kW], SoP_discharge [kW] from G2 · Fault severity from G3
OUTPUTS	CAN frame BMS_VCU_01 (CAN ID 0x111, 10 ms) to VCU · Contactor permission signal to G3 / A5
CONTROLS	ISO 11898-1 · OEM DBC signal encoding · $\pm 1\%$ power accuracy (CN-019) · AUTOSAR E2E Profile 2 CRC
MECHANISMS	AUTOSAR COM signal packing · SPC5748G CAN channel A · NvM for last-known-good values

4.5 A3 — Charging Interface Control

This sub-function manages both AC Level-2 and DC fast-charge sessions. It negotiates voltage and current limits with the EVSE and commands Group 3 contactors for charge enable/disable. Satisfies BMS-SYS-023.

ICOM	Content
INPUTS	SOC, max cell voltage, $T_{\max} \leq 45\text{ }^{\circ}\text{C}$ from G1/G2 · EVSE capability (max V, max I) from EVSE CAN
OUTPUTS	Charging voltage target [V], current limit [A] to EVSE · Charge enable/stop command to G3 contactors ($\leq 100\text{ ms}$) · Charging status (% , ETA, mode) to dashboard
CONTROLS	IEC 61851-1 charging standard · Overcharge cutoff at 100% SOC $\pm 1\%$ · Temperature limit $\leq 45\text{ }^{\circ}\text{C}$ · CCS/CHAdemo
MECHANISMS	DIN 70121 / ISO 15118 CAN stack · AUTOSAR NvM for charge session params · SPC5748G CAN channel B

4.6 A4 — Diagnostics and Data Logging

This sub-function implements the UDS ISO 14229 service handler, manages DTC storage in NvM with freeze-frame data, and maintains a 1 Hz ring-buffer data log. Satisfies BMS-SYS-019, BMS-SYS-024, and BMS-SYS-027.

ICOM	Content
INPUTS	Fault events + freeze-frame context from G3 · UDS request frames from tester (CAN ID 0x7DF / 0x7E4) · Periodic SOC, V, I, T at 1 Hz from G1/G2
OUTPUTS	UDS positive/negative response frames to tester ($\leq 2\text{ s}$) · DTC list + freeze frame to NvM (256 kB) · 1 Hz data log to internal flash
CONTROLS	ISO 14229 UDS · SAE J1979 OBD-II · DTC format SAE J2012 · Response $\leq 2\text{ s}$ (CN-024) · Coverage $\geq 95\%$
MECHANISMS	AUTOSAR Dcm (UDS server) + Dem (DTC manager) · 256 kB NvM EEPROM · CAN diagnostic address 0x7E4

4.7 A5 — Operating Mode and Wake-up Management

This sub-function implements the BEM finite state machine with 6 operating modes. It handles wake-up from Sleep in ≤ 500 ms, critical fault reaction in ≤ 100 ms, and carries the ASIL-C designation. Satisfies BMS-SYS-020, BMS-SYS-025, and BMS-SYS-026.

ICOM	Content
INPUTS	Wake trigger: CAN NM frame or KL15 · Fault severity (CRITICAL/WARNING/INFO) from G3 · Charge plug detection from A3 · VCU power request from A1
OUTPUTS	Active mode flag (BMS_Mode frame broadcast, 100 ms) · Power enable/disable to G3 contactors · Fault reaction command ≤ 100 ms to G3
CONTROLS	ISO 26262 ASIL-C FSM constraints · BMS-SYS-026 wake ≤ 500 ms · BMS-SYS-020 reaction ≤ 100 ms · BMS-SYS-025 mode list
MECHANISMS	AUTOSAR BswM mode manager · AUTOSAR CanNm network management · Hardware watchdog 10 ms · SPC5748G low-power mode

4.8 BEM Operating Mode State Machine

The six operating modes defined in BMS-SYS-025 are formally described with their entry conditions, CAN behavior during each mode, and exit triggers:

Mode	Code	Entry Condition	CAN / Comm Behavior	Exit Trigger
Initialization	0	KL15 ON or CAN NM wake frame	Self-test, NvM load. Bus silence until tests pass.	All checks pass → Normal (≤ 500 ms)
Normal Drive	1	Init OK + contactors closed	BMS_Status 10 ms, BMS_Energy 100 ms to all ECUs.	Fault / Plug detected / KL15 OFF
Charging	2	Plug detected + EVSE handshake OK	A3 active: V_target/I_limit negotiation. Dashboard ETA.	SOC = 100% or fault or unplug
Limp Mode	3	Non-critical fault (≤ 500 ms)	Reduced power flag on CAN. DTC logged. Driver warned.	Fault cleared → Normal; critical → Safe
Safe State	4	Critical fault (≤ 100 ms)	Contactors open. HV isolated. DTC + freeze	UDS reset by technician only

			frame. CAN silent except diagnostics.	
Sleep	5	KL15 OFF + CanNm go-to-sleep	CAN transceiver sleep. Min current draw. Watchdog active.	CAN NM frame or KL15 → Init ≤ 500 ms

4.9 Cross-Team Interface Definition

The following table defines all signals exchanged between Group 4 and the other groups, formalizing the integration boundaries:

Signal	Source	Destination	Interface	Cycle / Trigger
BMS_CellVoltage_Array	G1	G4 A1	BMS_InternalData_IF	100 ms
BMS_PackCurrent	G1	G4 A1	BMS_InternalData_IF	10 ms
BMS_TempMax/Min	G1	G4 A1/A3	BMS_InternalData_IF	100 ms
BMS_SOC	G2	G4 A2	BMS_InternalData_IF	100 ms
BMS_SoP_Charge/Discharge	G2	G4 A2	BMS_InternalData_IF	1 s
BMS_RangeEstimate	G2	G4 A1	BMS_InternalData_IF	30 s
BMS_FaultEvent	G3	G4 A4/A5	FaultEvent_IF	Event-driven
BMS_ContactorState	G3	G4 A5	Contactor_IF	10 ms
BMS_Mode_Command	G4 A5	G3	FaultReaction_IF	Event ≤ 100 ms

5. Formal Architecture Modeling — MATLAB System Composer

5.1 Overview

MATLAB System Composer is the MBSE tool used to formalize the SADT functional architecture into a machine-readable, requirement-traceable architectural model ready for Simulink behavioral simulation. The model is named BMS_System_Arch. Group 4's main component is BMS_CommModule, containing five sub-components derived from the SADT decomposition (A1–A5).

5.2 BMS_Profile — Stereotype Definitions

A custom profile BMS_Profile.xml was created containing seven stereotypes. Each stereotype encodes engineering requirements as structured metadata directly on the architecture model:

Stereotype	Base	Properties
ElectricalComponent	Component	supplyVoltage (V), powerConsumption_mW, operatingTempRange
Sensor	Component	samplingFrequency_Hz, accuracy, sensorType, redundancyLevel
Estimator	Component	updateRate_Hz, algorithmType, estimationError_pct, inputDependencies
SafetyFunction	Component	ASIL_Level (A/B/C/D), reactionTime_ms, faultCoverage_pct, safeState
Diagnostic	Component	DTC_prefix, diagnosticProtocol, testCoverage_pct
CANInterface	Port/Interface	CAN_ID (hex), messagePeriod_ms, DLC_bytes, signalEncoding, ASIL_Level
BMSModes	Component	Mode (enum: Shutdown/Standby/Init/Normal/Fault/Charging)

5.3 Sub-Component Definitions

Component	SAD T	Stereotype	Key Properties	Requirements
CANTxRxManager	A1	CANInterface	CAN_ID=0x110, period=10 ms, DLC=8, ASIL-C	SYS-021, SYS-033
VCUInterfaceHandler	A2	CANInterface	CAN_ID=0x111, period=10 ms, DLC=8, ASIL-C	SYS-022
ChargingInterfaceCtrl	A3	CANInterface	CAN_ID=0x120, period=100 ms, DLC=8, ASIL-B	SYS-023
DiagnosticsLogger	A4	Diagnostic	DTC_prefix=P0B00, protocol=UDS_ISO14229, coverage=95%	SYS-019, SYS-024, SYS-027
ModeWakeupManager	A5	SafetyFunction	ASIL-C, reactionTime=100 ms, coverage=99%, safeState=CONTACTORS_OPEN	SYS-020, SYS-025, SYS-026

5.4 Data Interface Definitions

Eighteen typed data interfaces were defined in the Interface Editor. Each interface specifies element names, data types (double/int/bool), and physical units. The key interfaces are:

Interface	Data Elements
BMS_VCU_IF	soc:double,% packVoltage:double,V packCurrent:double,A tempStatus:int,°C sopCharge:double,kW sopDischarge:double,kW
BMS_InternalData_IF	soc:double,% packVoltage:double,V packCurrent:double,A tempMax:double,°C tempMin:double,°C sopCharge:double,kW
FaultEvent_IF	faultID:int faultSeverity:int faultTimestamp:int,ms
ModeStatus_IF	activeMode:int (0=Init, 1=Normal, 2=Chg, 3=Limp, 4=Safe, 5=Sleep) modeTimestamp:int,ms
FaultReaction_IF	reactionCmd:int (0=ENABLE, 1=DISABLE, 2=SAFE) reactionLatency_ms:int,ms
ChargeParams_IF	voltageTarget:double,V currentLimit:double,A chargeEnable:bool
DTC_IF	dtcCode:int freezeFrameSOC:double,% freezeFrameVoltage:double,V freezeFrameTemp:double,°C
DataLog_IF	logSOC:double,% logVoltage:double,V logCurrent:double,A logTemp:double,°C logTimestamp:int,ms

Contactors_IF	contactorCmd:bool prechargeEnable:bool
Wake_IF	canNM_Wake:bool k115_signal:bool

5.5 Architectural Views

5.5.1 CAN Communication View

Filter: stereotype = CANInterface. Displays CANTxRxManager (0x110, 10 ms), VCUInterfaceHandler (0x111, 10 ms), and ChargingInterfaceCtrl (0x120, 100 ms) with all CAN_ID, messagePeriod_ms, DLC_bytes, and ASIL_Level properties visible. Intended audience: CAN network architect verifying message schedule consistency.

5.5.2 ASIL Safety View

Filter: stereotype = SafetyFunction. Displays ModeWakeupManager with ASIL_Level=C, reactionTime_ms=100, faultCoverage_pct=99, safeState=CONTACTORS_OPEN. This view immediately identifies that ModeWakeupManager is the sole ASIL-C block in Group 4. Intended audience: functional safety engineer performing ISO 26262 compliance review.

6. Simulink Behavioral Implementation

6.1 Overview

Each System Composer sub-component was linked to a Simulink subsystem via the 'Add Simulink Behavior' action. Port names and interface definitions from System Composer become the Inport and Outport names in Simulink, and Bus Objects corresponding to each interface are auto-generated. The key Simulink blocks used across all sub-components are:

Bus Selector	Extracts named signals from a structured bus — equivalent to struct field access in C. Used at the input of every sub-component to unpack incoming buses.
Bus Creator	Packages individual signals into a structured bus — equivalent to struct construction. Used at the output of every sub-component to assemble outgoing buses.
Relational Operator	Compares two scalar signals, outputs a boolean. Used to implement safety threshold checks ($SOC < 100\%$, $T < 45^{\circ}C$, $faultSeverity < 2$).
Logical Operator (AND)	Logical AND of boolean inputs. Used in A3 to combine dual charging safety conditions into a single chargeEnable signal.
Constant	Fixed scalar value. Used to encode requirement thresholds directly in the model (CAN IDs, DLC, timing budgets, temperature limits).
Clock	Current simulation time in seconds. Used in A4 to timestamp data log entries and in A5 for mode timing guards.
Stateflow Chart	Formal finite state machine with guarded transitions and entry/exit actions. Used exclusively in A5 (ModeWakeupManager) to implement the 6-mode BEM FSM.

6.2 A1 — CANTxRxManager

Output: out_CAN_TxFrames (BMS-SYS-021)

A Bus Selector on in_BMSData extracts soc, packVoltage, packCurrent, tempMax, sopCharge, and sopDischarge. A Bus Creator assembles the CAN_TxFrame_IF structure with four fields: frameID = Constant 272 (= 0x110 hex), frameData from the Bus Selector output, dlc = Constant

8, and txPeriod_ms = Constant 10. These constants directly encode requirement BMS-SYS-021 (10 ms cycle, CAN ID 0x110, 8-byte DLC) as verifiable model parameters.

Output: out_ParsedVCURequest

A Bus Selector on in_VCU_RxFrame extracts frameData (mapped to powerRequest_kW) and frameValid (mapped to requestValid). A Bus Creator assembles VCU_Request_IF. This parsed request is forwarded to A2 for power context and to A5 for mode management.

Output: out_ParsedEVSECap

A Bus Selector on in_EVSE_RxFrame extracts frameData to maxVoltage_V. maxCurrent_A is set to Constant 250 A in the simplified model. A Bus Creator assembles EVSE_Cap_IF and forwards it to A3.

6.3 A2 — VCUInterfaceHandler

Output: out_VCU_CAN_Frame (BMS-SYS-022)

A Bus Selector on in_BMSData extracts six signals: soc, packVoltage, packCurrent, tempMax, sopCharge, and sopDischarge. A Bus Creator with six named inputs assembles BMS_VCU_IF directly to out_VCU_CAN_Frame. This satisfies the complete signal set required by BMS-SYS-022.

Output: out_ContactorPermission

A Bus Selector on in_FaultSeverity extracts faultSeverity. A Relational Operator (operator '<', threshold Constant 2) outputs true when faultSeverity < 2 — that is, when no critical fault is present. This encodes the three-level severity classification: 0 = Info, 1 = Warning, 2 = Critical. prechargeEnable is set to Constant 0. A Bus Creator assembles Contactor_IF.

6.4 A3 — ChargingInterfaceCtrl

Charging Authorization Logic (BMS-SYS-023)

Two parallel safety conditions are evaluated simultaneously using Relational Operator blocks:

- Condition 1 — SOC check: Bus Selector extracts soc → Relational Operator '<' → Constant 100 → soc_ok = true when SOC < 100%.
- Condition 2 — Temperature check: Bus Selector extracts tempMax → Relational Operator '<' → Constant 45 → temp_ok = true when T_max < 45 °C.
- A Logical Operator (AND) combines soc_ok and temp_ok into chargeEnable. Charging is authorized only when the battery is not full and not thermally stressed simultaneously.

Outputs: out_ChargeParams, out_ChargeStatus, out_ChargePlug

A Bus Selector on in_EVSE_Capability extracts maxVoltage_V and maxCurrent_A. A Bus Creator assembles ChargeParams_IF with voltageTarget, currentLimit, and chargeEnable from the AND gate. ChargeStatus_IF is assembled with chargePct from soc, eta_min from Constant 0 (simplified), and chargeMode from Constant 1 (DC charging). ChargePlug_IF is assembled with plugDetected from chargeEnable and evseHandshake from Constant 1, and is sent to A5 to trigger the CHARGING mode transition.

6.5 A4 — DiagnosticsLogger**Output: out_DTC_Store (BMS-SYS-019)**

Bus Selectors unpack in_FaultEvent (extracting faultID, faultSeverity, faultTimestamp) and in_BMSData (extracting soc, packVoltage, tempMax). A Bus Creator assembles DTC_IF: dtcCode from faultID, freezeFrameSOC from soc, freezeFrameVoltage from packVoltage, and freezeFrameTemp from tempMax. This freeze-frame approach captures the complete system state at the moment of fault occurrence, consistent with OBD-II standard practice.

Output: out_UDS_Response (BMS-SYS-024)

A Bus Selector on in_UDS_Request extracts requestData. A Bus Creator assembles UDS_Response_IF with responseCanID from Constant 2020 (= 0x7E4, the standard physical diagnostic address for BMS per ISO 15765), responseData from requestData (echo response, simplified), and responseValid from Constant 1. A production implementation would include the full UDS service handler (ReadDTCInformation, ClearDTC, ReadDataByIdentifier).

Output: out_DataLog (BMS-SYS-027)

A Bus Selector on in_BMSData extracts soc, packVoltage, packCurrent, and tempMax. A Clock block from Simulink Sources provides the current simulation time as logTimestamp. A Bus Creator assembles DataLog_IF with all five fields. The Clock block represents a hardware RTC timer in the production ECU, providing the 1 Hz timestamped ring buffer required by BMS-SYS-027.

6.6 A5 — ModeWakeupManager (Stateflow)**6.6.1 Input Processing**

Five Bus Selector blocks unpack the five input buses before feeding the Stateflow Chart: canNM_Wake and kl15_signal from in_WakeTrigger; faultSeverity from in_FaultClassification; plugDetected from in_ChargePlug; powerRequest_kW from in_VCU_PowerReq; and contactorCmd from in_ContactorPermission.

6.6.2 Stateflow States and Entry Actions

State	activeMode	contactorCmd	reactionCmd	Entry Action
INIT	0	false	0	activeMode=0; contactorCmd=false; reactionCmd=0;
NORMAL	1	true	0	activeMode=1; contactorCmd=true; reactionCmd=0;
CHARGING	2	true	0	activeMode=2; contactorCmd=true; reactionCmd=0;
LIMP	3	true	1	activeMode=3; contactorCmd=true; reactionCmd=1;
SAFE_STATE	4	false	2	activeMode=4; contactorCmd=false; reactionCmd=2;
SLEEP	5	false	0	activeMode=5; contactorCmd=false; reactionCmd=0;

6.6.3 Stateflow Transitions

From	To	Guard Condition	Requirement
INIT	NORMAL	[after(0.5, sec)]	BMS-SYS-026 (≤ 500 ms init)
NORMAL	CHARGING	[plugDetected == true]	BMS-SYS-025
NORMAL	LIMP	[faultSeverity == 1]	BMS-SYS-025
NORMAL	SAFE_STATE	[faultSeverity == 2]	BMS-SYS-020 (≤ 100 ms)
NORMAL	SLEEP	[kl15_signal == false]	BMS-SYS-025
SLEEP	INIT	[canNM_Wake==true kl15_signal==true]	BMS-SYS-026
CHARGING	NORMAL	[plugDetected == false]	BMS-SYS-025
LIMP	SAFE_STATE	[faultSeverity == 2]	BMS-SYS-020
SAFE_STATE	INIT	[after(5, sec)]	Simplified reset (UDS session in production)

6.6.4 Output Bus Assembly

Three Bus Creators package the Stateflow outputs: out_ActiveMode bundles activeMode with modeTimestamp from Clock; out_PowerEnable bundles contactorCmd with prechargeEnable = Constant 0; out_FaultReaction bundles reactionCmd with reactionLatency_ms = Constant 100 —

formally encoding the 100 ms ASIL-C reaction time budget of BMS-SYS-020 as a verifiable model parameter

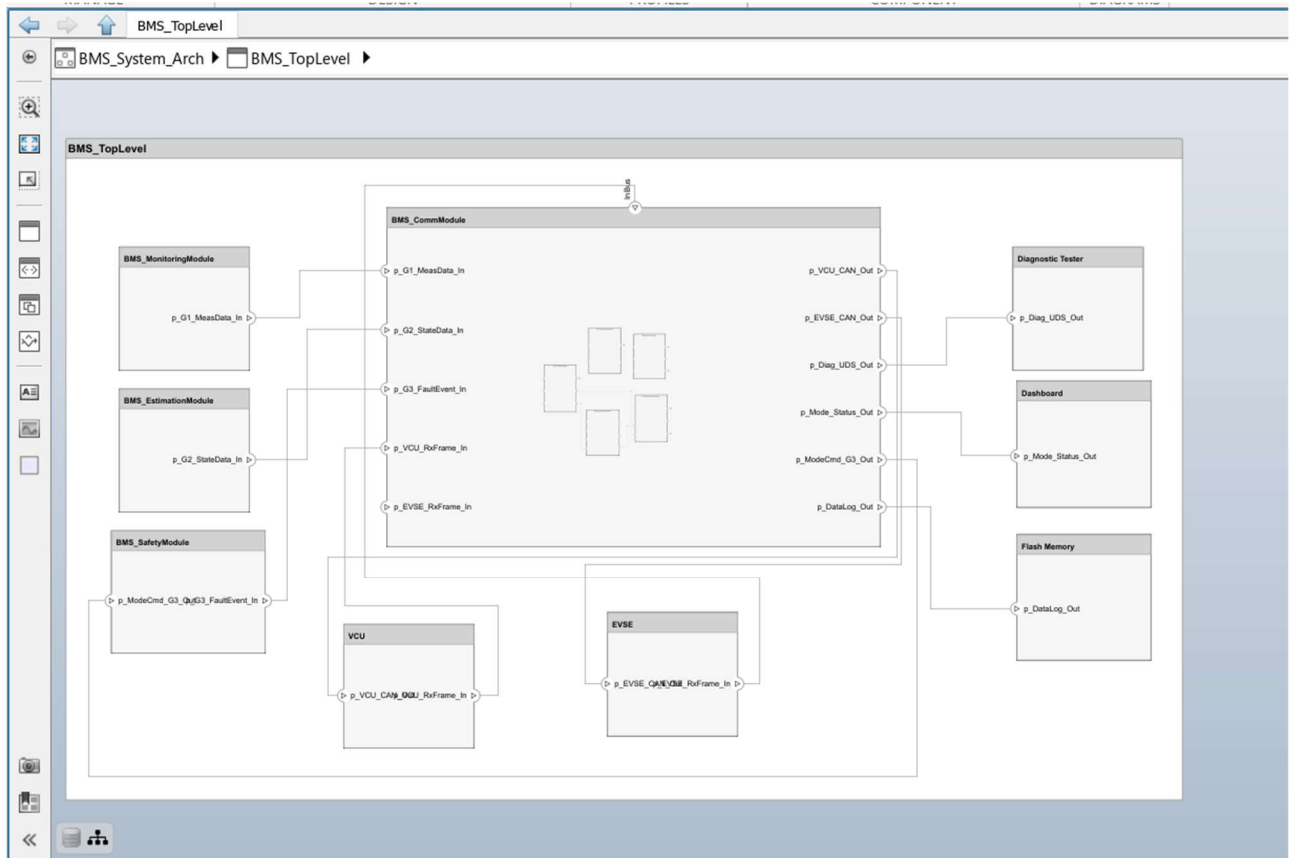


Figure 1: Top-Level Architecture of the Battery Management System (BMS)

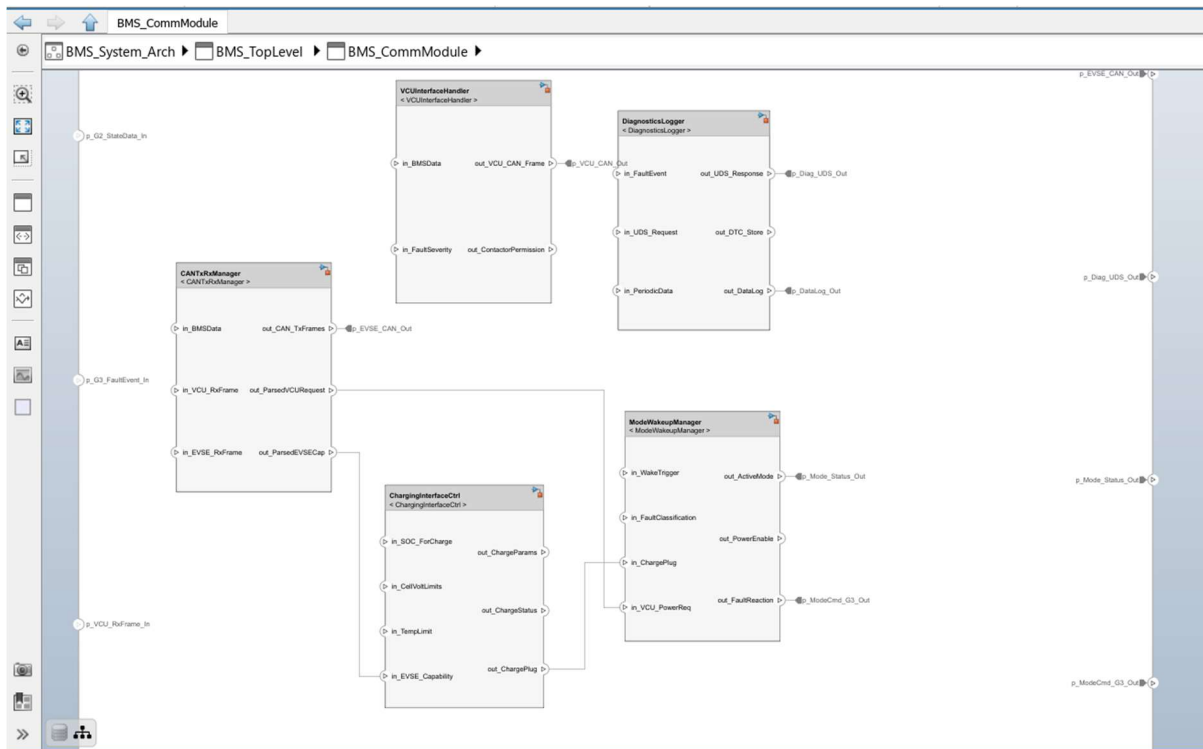


Figure 2: Functional Architecture of the BMS Communication Module

7. CAN Communication Architecture

7.1 Network Overview

The BEM communication system uses CAN 2.0B at 500 kbps. The network includes the BMS, the VCU (master ECU), the EVSE charger, the dashboard instrument cluster, the diagnostic tester (OBD-II connector), and the thermal ECU. Message priority is determined by CAN arbitration: lower CAN ID = higher bus priority.

7.2 Message Schedule

CAN ID	Message Name	Period	DLC	Key Signals	Requirement
0x110 (272)	BMS_Status	10 ms	8 bytes	SOC, V_pack, I_pack, T_max — safety-critical, cyclic	SYS-021
0x111 (273)	BMS_VCU_01	10 ms	8 bytes	soc, packVoltage, packCurrent, tempStatus, sopCharge, sopDischarge	SYS-022
0x120 (288)	BMS_Charging	100 ms	8 bytes	voltageTarget, currentLimit, chargeEnable — to EVSE	SYS-023
0x7E4 (2020)	UDS_Response	On request	8 bytes	responseCanID, responseData, responseValid — UDS physical	SYS-024
0x7DF (2015)	UDS_Request_Func	On request	8 bytes	canID, serviceID, requestData — UDS functional address	SYS-024
BMS_Mode	BMS_Mode_Status	100 ms	4 bytes	activeMode (0–5), modeTimestamp — broadcast to all ECUs	SYS-025

7.3 Signal Encoding and Diagnostic Communication

All safety-critical CAN frames (BMS_VCU_01 and BMS_Status) use the AUTOSAR E2E Profile 2 CRC scheme for end-to-end protection against transmission errors. Charging interface signals use IEC 61851 encoding conventions. All signals use Intel byte order (little-endian) with a fixed DLC of 8 bytes.

The UDS diagnostic interface uses physical address 0x7E4 (tester to BMS) and functional broadcast address 0x7DF. Transport protocol follows ISO 15765-2. Implemented services include ReadDTCInformation (0x19), ClearDTCInformation (0x14), and ReadDataByIdentifier (0x22). The DTC prefix is P0B00 (powertrain, BMS-specific range). Response time is ≤ 2000 ms per BMS-SYS-024.

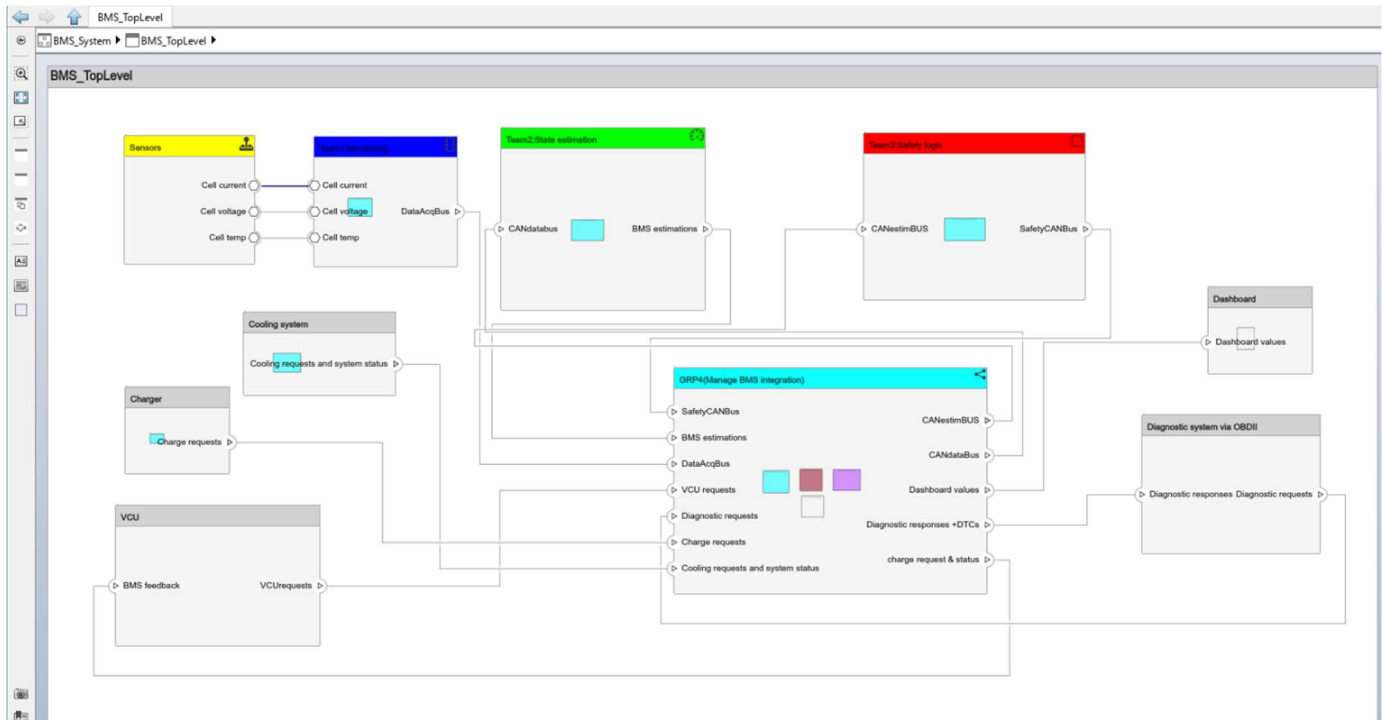


Figure 3: Top-Level Architecture of the Battery Management System (BMS)

8. Safety Analysis and ISO 26262 Compliance

8.1 ASIL-C Rationale for ModeWakeupManager

ISO 26262 assigns Automotive Safety Integrity Levels (ASIL A to D) based on three parameters: Severity (S0–S3), Exposure (E0–E4), and Controllability (C0–C3). The operating mode management function (A5) carries ASIL-C because a failure to enter Safe State within 100 ms of a critical fault — for example, a cell overvoltage event — could result in uncontrolled current flow through the 355 V pack, potentially leading to thermal runaway, fire, or explosion (S3, E4, C2 → ASIL-C).

Property	Value	Justification
ASIL_Level	C	Fault in mode management → uncontrolled HV discharge → fire risk (S3, E4, C2)
reactionTime_ms	100	Fault detection (G3, ~20 ms) + CAN TX (~5 ms) + contactor actuation (~30 ms) ≤ 100 ms total
faultCoverage_pct	99	99% of fault modes detectable: overvoltage, undervoltage, overcurrent, overtemperature, short circuit, comm. timeout
safeState	CONTACTORS_OPEN	Opening main contactors isolates the 355 V pack from the vehicle — the definitive safe state for any HV battery fault

8.2 Safety Layers

Group 4 implements a five-layer safety architecture to meet the ASIL-C requirements:

- Layer 1 — AUTOSAR E2E protection: CRC on all safety-critical CAN frames prevents undetected message corruption during transmission.
- Layer 2 — Stateflow formal FSM: a deterministic state machine with guarded transitions prevents undefined or illegal mode behavior.
- Layer 3 — Hardware watchdog (10 ms): if the software fails to respond within 10 ms, the hardware automatically opens the main contactors.
- Layer 4 — Fault severity classification: a three-level severity system (Info/Warning/Critical) enables a graduated response proportional to actual risk.
- Layer 5 — Reaction time decomposition: the 100 ms budget is formally allocated across fault detection (G3), CAN transmission (A1), and contactor actuation (G3), ensuring each phase stays within its timing budget.

8.3 Critical Fault Reaction Sequence

When Group 3 detects a critical fault (e.g., cell overvoltage > 4.2 V), the following sequence executes within the 100 ms ASIL-C budget:

- G3 classifies the fault as `faultSeverity = 2` (Critical) and transmits `FaultEvent_IF` over `p_G3_FaultEvent_In`.
- A4 (DiagnosticsLogger) receives the fault event and logs DTC code P0B00 with a complete freeze frame (SOC, V, T at fault time) to NvM flash.
- A5 (ModeWakeupManager) receives `faultSeverity = 2`, and the Stateflow chart transitions `NORMAL → SAFE_STATE`, setting `contactorCmd = false` and `reactionCmd = 2`.
- A5 transmits `out_FaultReaction` (`reactionCmd=2`, `reactionLatency_ms=100`) to G3 via `p_ModeCmd_G3_Out`.
- G3 opens the main contactors, isolating the 355 V pack from the vehicle electrical system.
- A1 broadcasts `BMS_Mode_Status` with `activeMode = 4` (Safe State) to all network ECUs at the next 100 ms cycle.
- The system remains in Safe State until a technician initiates a UDS diagnostic session with `SecurityAccess + ECUReset`. No automatic recovery is possible.

9. Requirements Traceability

9.1 Traceability Method

All 33 BEM system requirements were imported into System Composer from Proposed_RTM.xlsx via Requirements → Import → From Spreadsheet. The imported file was saved as BEM_Requirements.slreqx. Each Group 4 requirement was linked to its responsible component using 'Link to Selected Element', creating formal bidirectional traceability from the RTM to the architecture, and from the architecture to Simulink constants and Stateflow guard conditions.

9.2 Traceability Matrix — Group 4

Req ID	Title	Component	Stereotype Encoding	Simulink Encoding
BMS-SYS-019	DTC Logging	DiagnosticsLogger	DTC_prefix=P0B00, protocol=UDS_ISO14229	Bus Creator → out_DTC_Store with freeze frame fields
BMS-SYS-020	Fault Reaction	ModeWakeupManager	ASIL_Level=C, reactionTime_ms=100	Stateflow: NORMAL→SAFE_STATE [faultSeverity==2]; Constant 100 in out_FaultReaction
BMS-SYS-021	CAN Comm.	CANTxRxManager	CAN_ID=0x110, period=10 ms, DLC=8	Constant 272, Constant 10, Constant 8 in Bus Creator → out_CAN_TxFrames
BMS-SYS-022	VCU Interface	VCUInterfaceHandler	CAN_ID=0x111, period=10 ms	Bus Creator → out_VCU_CAN_Frame with 6 BMS signals
BMS-SYS-023	Charging IF	ChargingInterfaceCtrl	CAN_ID=0x120, period=100 ms	Relational Operators (SOC<100, T<45) + AND → chargeEnable
BMS-SYS-024	UDS Diagnostics	DiagnosticsLogger	protocol=UDS_ISO14229, coverage=95%	Constant 2020 (=0x7E4) → responseCanID in out_UDS_Response
BMS-SYS-025	Op. Modes	ModeWakeupManager	safeState=CONTACTORS_OPEN	Stateflow 6-state FSM with guarded transitions
BMS-SYS-026	Wake-up	ModeWakeupManager	reactionTime_ms=100	Stateflow: SLEEP→INIT [canNM_Wake[kl15]]; INIT→NORMAL after(0.5, sec)
BMS-SYS-027	Data Log	DiagnosticsLogger	testCoverage_pct=95	Clock block → logTimestamp in Bus Creator → out_DataLog

10. Engineering Challenges and Resolutions

10.1 System Composer Model Issues

Issue	Root Cause	Resolution
Architecture model contains no components	Sub-components had no Simulink behavior attached; export produced empty subsystems.	Applied 'Add Simulink Behavior' to all 5 sub-components, BMS_CommModule, and BMS_TopLevel. Verified Simulink icon appeared before re-exporting via File → Export → Overwrite.
Bus hierarchy mismatch at boundary port	Output port had no interface assigned; generated Bus Object expected structured signal but received untyped signal.	Assigned correct interface to every output port in Property Inspector → Interface field. Re-exported and ran Update Model (Ctrl+D) to regenerate all Bus Objects.
ChargePlug_IF name mismatch at in_ChargePlug	A3 output (Contactor_IF: contactorCmd/prechargeEnable) was incorrectly wired to A5 input (ChargePlug_IF: plugDetected/evseHandshake).	Created dedicated out_ChargePlug port on A3 with ChargePlug_IF. Wired plugDetected from chargeEnable AND gate, evseHandshake from Constant 1. Reconnected A3→A5.
in_BMSData missing in DiagnosticsLogger	The port was added in System Composer after the first export and was absent from the Simulink subsystem.	Added new Inport block named in_BMSData with interface BMS_InternalData_IF in the subsystem. Re-connected to Bus Selector for log data extraction.

10.2 Profile Definition Challenges

The BMS_Profile.xml required careful specification of property types, default values, ranges, and units for each stereotype. Key challenges included: the BMSModes enumeration required a separate MATLAB class file (BMSModes.m) inheriting from Simulink.IntEnumType; float property types required explicit unit declarations matching the System Composer unit registry; and stereotype default mapping needed correct configuration to ensure new components automatically received the appropriate stereotype. The solution was to follow the System Composer Profile Editor workflow precisely and validate the profile against a test model before applying it to the main architecture.

10.3 SADT-to-Model Traceability

Maintaining traceability between the SADT functional decomposition and the System Composer architecture model required careful cross-referencing throughout the project. The ChargePlug interface restructuring — moving from a shared Contactor_IF to a dedicated ChargePlug_IF between A3 and A5 — required updates to the cross-team interface table, the boundary port

definitions, and the RTM. The solution was to treat the RTM as a living document and update all affected artifacts synchronously whenever an architectural decision changed.

11. Model Limitations and Future Improvements

11.1 Current Limitations

The current Simulink implementation is a functional-level model. The following simplifications were made consciously and are documented here for future development teams:

- CAN bus simulation: the model does not simulate CAN bus load, message arbitration, transmission errors, or bus-off recovery. Only signal values are modeled.
- UDS service handler: A4 implements only a simplified echo response. Production-grade implementation requires full ISO 14229 service processing including session management, security access, and DTC aging.
- EVSE interface: A3 uses fixed Constant values for maxVoltage_V and maxCurrent_A. Production model would parse these from real EVSE CAN messages per ISO 15118.
- Safe State recovery: A5 uses after(5, sec) as the recovery trigger. Real system recovery requires a technician UDS session (SecurityAccess + ECUReset).
- AUTOSAR E2E CRC: the CRC is represented as a Constant in stereotype properties but is not computed in the Simulink model.

11.2 Proposed Future Improvements

- Integrate the MATLAB Vehicle Network Toolbox to simulate CAN bus load, message timing, and error handling under realistic network conditions including bus-off scenarios.
- Implement the full UDS service handler with session state machine, security access (0x27), DTC aging counter management, and multi-frame ISO 15765-2 transport.
- Replace simplified EVSE Constant blocks with a CAN RX parser extracting maxVoltage_V and maxCurrent_A from real EVSE CAN frames per ISO 15118 / DIN 70121.
- Add Hardware-in-the-Loop (HIL) testing capability to validate BMS software against real CAN hardware interfaces on the SPC5748G target.
- Implement AUTOSAR-compliant software component descriptions to enable direct code generation using Embedded Coder for the SPC5748G microcontroller.
- Extend the Stateflow FSM with full ISO 26262 ASIL-C requirements: safe state confirmation feedback, redundant fault detection, and latent fault monitoring.

12. Conclusion

This report has presented the complete systems engineering lifecycle of the BEM Communication and System Integration domain developed by Group 4 at ENISo. The work progressed systematically through five engineering disciplines — system context analysis, requirements formalization, functional architecture decomposition, formal MBSE modeling, and Simulink behavioral implementation — producing a coherent, traceable, and validated engineering artifact.

The target vehicle's 78.1 kWh, 96s46p lithium-ion battery pack and its surrounding vehicle ecosystem established the engineering context. From this context, nine primary system requirements (BMS-SYS-019 to BMS-SYS-027) were derived and formalized with explicit verification methods. The SADT/IDEF0 decomposition translated these requirements into five functional sub-activities (A1–A5), each with complete ICOM tables and cross-team interface definitions. MATLAB System Composer then formalized this architecture with five sub-components, seven stereotypes, eighteen typed data interfaces, and two architectural views. Finally, Simulink implemented the behavioral logic using Bus Selectors, Bus Creators, Relational Operators, Logical AND gates, and a Stateflow FSM for the ASIL-C operating mode manager.

The key technical contributions of the project are: a formally validated System Composer architecture with full stereotype annotations and requirement traceability; a Stateflow-based ASIL-C operating mode FSM with 6 states, 9 guarded transitions, and direct requirement encoding; a dual-condition charging safety logic ($SOC < 100\%$ AND $T < 45^{\circ}C$); and full bidirectional traceability from customer needs through system requirements to architecture components to Simulink model parameters.

The architecture model, profile, and Simulink simulation provide a solid foundation for integration testing with the complete BEM system across all four groups within the shared BMS_System_Arch model, and for future progression toward AUTOSAR code generation and Hardware-in-the-Loop validation.

13. References :

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